

## 1. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1. Data type of all columns in the "customers" table.

```
select
  column_name,
  data_type
from
  scaler123-459505.target.INFORMATION_SCHEMA.COLUMNS
where table_name = 'customers';
```

| Row | column_name ▼            | data_type ▼ |
|-----|--------------------------|-------------|
| 1   | customer_id              | STRING      |
| 2   | customer_unique_id       | STRING      |
| 3   | customer_zip_code_prefix | INT64       |
| 4   | customer_city            | STRING      |
| 5   | customer_state           | STRING      |

### Insight:

1. Customers table include customer id, customer unique id, customer zip code prefix, customer city, customer state.
2. Get the time range between which the orders were placed.

```
SELECT
  max(order_purchase_timestamp) as max_time,
  min(order_purchase_timestamp) as min_time
FROM target.orders;
```

| Row | max_time ▼              | min_time ▼              |
|-----|-------------------------|-------------------------|
| 1   | 2018-10-17 17:30:18 UTC | 2016-09-04 21:15:19 UTC |

### Insights:

1. Max time when the order placed is 5:30 in evening.
2. Min time when the order placed is 9:15 in night.
3. Count the Cities & States of customers who ordered during the given period.

```
SELECT count(DISTINCT c.customer_city) as city_cnt,
count(distinct c.customer_state) as state_cnt FROM target.customers c INNER
```

```
JOIN target.orders o on c.customer_id=o.customer_id;
```

| Row | city_cnt | state_cnt |
|-----|----------|-----------|
| 1   | 4119     | 27        |

### Insights:

1. Total city are 4119.
2. Total states are 27

## 2. In-depth Exploration:

1. Is there a growing trend in the no. of orders placed over the past years?

```
SELECT extract(YEAR from order_purchase_timestamp) as order_year,  
count(order_id) as total_order FROM target.orders group by order_year  
order by order_year;
```

| Row | order_year | total_order |
|-----|------------|-------------|
| 1   | 2016       | 329         |
| 2   | 2017       | 45101       |
| 3   | 2018       | 54011       |

### Insights:

1. In 2016, total order are only 329.
2. In 2017, total order increases from 329 to a large profit 45101.
3. In 2018, total order increases to 20% i.e. 54011 from 2017.

### Recommendation:

1. Every year number of orders are increasing so, the company make sure to handle customers, storage and staff efficiently.
  2. After 2016, sudden increase in orders which is beneficial for company so to keep this up start doing advertisements, promotion and give discount to customers.
  3. Keep feedback system for the customers to ensure that the orders are going better.
2. Can we see some kind of monthly seasonality in terms of the no. of orders being placed?

```
SELECT extract(month from order_purchase_timestamp) as order_month,
count(order_id) as total_order FROM target.orders group by order_month order
by order_month;
```

| Row | order_month ▼ | total_order ▼ |
|-----|---------------|---------------|
| 1   | 1             | 8069          |
| 2   | 2             | 8508          |
| 3   | 3             | 9893          |
| 4   | 4             | 9343          |
| 5   | 5             | 10573         |
| 6   | 6             | 9412          |
| 7   | 7             | 10318         |
| 8   | 8             | 10843         |
| 9   | 9             | 4305          |
| 10  | 10            | 4959          |
| 11  | 11            | 7544          |
| 12  | 12            | 5674          |

### Insights:

1. From January to August, we can see the rising trend in the orders placed.
2. After August there is a huge difference found in orders i.e. 10843 to 4305 and it is the lowest among all.
3. Orders started recovering October to November but again the difference comes from 7544 to 5674.
4. Peak order month is August .
5. Lowest order month is September.
6. We can also see that rainy season the sales is high.

### Recommendations:

1. To maintain the high demand in first eight month company should provide sufficient inventory and staffs ready.
2. Investigate reasons for the decrease in September and December, it could be seasonal or operational.
3. In September and December, start more advertisement ,campaigns and promotions to boost sales.
4. Use May ,July and August as the benchmark to set target to other months.

5. Use festive or year end campaigns in December to boost lower order volume.
6. During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)
  - a. 0-6 hrs : Dawn
  - b. 7-12 hrs : Mornings
  - c. 13-18 hrs : Afternoon
  - d. 19-23 hrs : Night

```
SELECT
CASE
  WHEN extract(HOUR from order_purchase_timestamp) between 0 and 6 THEN
    'Dawn'
  WHEN extract(HOUR from order_purchase_timestamp) between 7 and 12 THEN
    'Morning'
  WHEN extract(HOUR from order_purchase_timestamp) between 13 and 18 THEN
    'Afternoon'
  ELSE 'Night'
END as time_of_day,
count(order_id) as total_orders
FROM target.orders
group by time_of_day
order by total_orders desc;
```

| Row | time_of_day ▼ | total_orders ▼ |
|-----|---------------|----------------|
| 1   | Afternoon     | 38135          |
| 2   | Night         | 28331          |
| 3   | Morning       | 27733          |
| 4   | Dawn          | 5242           |

### Insights:

1. Afternoon has the highest number of orders i.e. 38135.
2. Night and Morning are close in terms of total orders from 28331 to 27733.
3. Dawn has the lowest order volume i.e. 5242 that shows the user activity is very low.

### Recommendation:

1. Since most orders happen in the afternoon, we should focus on giving discount during that time.
2. We can go for some deals in night and morning also because the orders are quite good at that time.
3. Try avoiding dawn because very few people are active that time.
4. We should cover mostly from Afternoon, Night and Morning only.

5. Use proper app for the orders to handle peak orders smoothly.

### 3. Evolution of E-commerce orders in the Brazil region:

1. Get the month on month no. of orders placed in each state.

```
select customer_state, a.month, count(order_id) as total_order
from (select * , extract(month from order_purchase_timestamp) as month from
target.orders inner join target.customers using(customer_id)) a
group by 1,2
order by 1,2;
```

| Row | customer_state ▼ | month ▼ | total_order ▼ |
|-----|------------------|---------|---------------|
| 1   | AC               | 1       | 8             |
| 2   | AC               | 2       | 6             |
| 3   | AC               | 3       | 4             |
| 4   | AC               | 4       | 9             |
| 5   | AC               | 5       | 10            |
| 6   | AC               | 6       | 7             |
| 7   | AC               | 7       | 9             |
| 8   | AC               | 8       | 7             |
| 9   | AC               | 9       | 5             |
| 10  | AC               | 10      | 6             |

#### Insights:

1. We can see the month on month orders placed in every state.
2. We can see the state AC with there highest orders month is May with 10 orders.
3. March has the lowest orders of 4.

#### Recommendations:

1. For states with high orders, keep promoting products regularly and offer good delivery services to keep customer happy.
2. The states with lower orders, try some special offers or discount to attract more people.
3. Keep changing data every few months to see changes and update yoyr plans.

2. How are the customers distributed across all the states?

```
SELECT customer_state, count(DISTINCT customer_id) as total_customer
FROM target.customers
group by customer_state
order by total_customer desc;
```

| Row | customer_state ▼ | total_customer ▼ |
|-----|------------------|------------------|
| 1   | SP               | 41746            |
| 2   | RJ               | 12852            |
| 3   | MG               | 11635            |
| 4   | RS               | 5466             |
| 5   | PR               | 5045             |
| 6   | SC               | 3637             |
| 7   | BA               | 3380             |
| 8   | DF               | 2140             |
| 9   | ES               | 2033             |
| 10  | GO               | 2020             |

### Insights:

1. SP has the highest number of customers with 41,746.
2. RJ and MG has the next high customers with 12852 and 11635 customers respectively.
3. After that we can see the sudden fall of customers .
4. States like GO ,DF , and ES has very few customers around 2000.

### Recommendation:

1. Focus more on GO, ES and DF as they have the lowest customers.
2. SP, RJ, and MG should follow their trends as it was going for the same growth.
3. Try welcome discount in low ones.
4. Premium services for the top states which has most customers.

## 4. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

1. Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only).

```
(
    with extract_order as
```

```

select extract(month from order_purchase_timestamp) as order_month,
extract(year from order_purchase_timestamp) as order_year,
o.order_id
from target.orders o
where extract(month from order_purchase_timestamp) between 1 and 8 and
extract(year from order_purchase_timestamp) in (2017,2018)),
cost_order as (
select sum(p.payment_value) as total_payment, e.order_year from target.payments p
join extract_order e on e.order_id=p.order_id
group by e.order_year
)
SELECT
MAX(CASE WHEN order_year = 2017 THEN total_payment END) AS total_2017,
MAX(CASE WHEN order_year = 2018 THEN total_payment END) AS total_2018,
ROUND(
100.0 * (
MAX(CASE WHEN order_year = 2018 THEN total_payment END) -
MAX(CASE WHEN order_year = 2017 THEN total_payment END)
) / IFNULL(MAX(CASE WHEN order_year = 2017 THEN total_payment END), 0),
2
) AS percent_increase
FROM
cost_order;

```

| Row | total_2017        | total_2018        | percent_increase |
|-----|-------------------|-------------------|------------------|
| 1   | 3669022.120000... | 8694733.840000... | 136.98           |

### Insights:

1. There was a big jump in order spending from 2017 to 2018.
2. The total amount spent went from ₹36.7 lakhs to ₹86.9 lakhs.
3. This is a sharp rise of nearly 137%.
4. It shows more customers are buying online or spending more per order.
5. E-commerce is growing fast and becoming more popular

### Recommendation:

1. This is a great time to start selling online as the market is expanding.
  2. Make sure your products are good and delivered on time.
  3. Use social media to reach more customers and build trust.
  4. Keep track of what products are trending or in demand.
  5. Provide simple payment methods like UPI, COD, or EMI.
2. Calculate the Total & Average value of order price for each state.

```

SELECT c.customer_state, round(sum(oi.price),2) as total_order_price,
round(avg(oi.price),2) as avg_order_price

```

```
FROM target.customers c join target.orders o on c.customer_id=o.customer_id
join target.order_items oi on o.order_id=oi.order_id
group by c.customer_state
order by 2 desc;
```

| Row | customer_state ▼ | total_order_price ▼ | avg_order_price ▼ |
|-----|------------------|---------------------|-------------------|
| 1   | SP               | 5202955.05          | 109.65            |
| 2   | RJ               | 1824092.67          | 125.12            |
| 3   | MG               | 1585308.03          | 120.75            |
| 4   | RS               | 750304.02           | 120.34            |
| 5   | PR               | 683083.76           | 119.0             |
| 6   | SC               | 520553.34           | 124.65            |
| 7   | BA               | 511349.99           | 134.6             |
| 8   | DF               | 302603.94           | 125.77            |
| 9   | GO               | 294591.95           | 126.27            |
| 10  | ES               | 275037.31           | 121.91            |

### Insights:

1. SP has the highest total order value — over 5202955, which means more people or more frequent purchases are made there.
2. BA has the highest average order price — around 134.6, even though its total spending is not the highest. This means customers here spend more per order.
3. RJ and MG also show high total order values and good average prices, making them strong markets too.
4. RS, PR, and SC show a balance of decent total sales and average price.

### Recommendation:

1. Since it brings the highest sales, keep investing in ads, faster delivery, and stock availability in this region.
2. People here spend more on each order. Promote luxury or high-quality products in this state.
3. DF, GO, and ES have smaller total sales but good average order values. These could grow fast with the right offers or service improvements.
4. Use this state-wise data to send custom ads or product suggestions based on local buying behavior.
5. Try comparing these order values with freight charges to see where profits are higher or where shipping is expensive

3. Calculate the Total & Average value of order freight for each state.



```

select c.customer_state, round(sum(oi.freight_value),2) as
total_freight_value, round(avg(oi.freight_value),2) as avg_freight_value
from target.customers c join target.orders o on c.customer_id=o.customer_id
join target.order_items oi on o.order_id=oi.order_id
where o.order_status = 'delivered'
group by c.customer_state
order by 2 desc;

```

| Row | customer_state | total_freight_value | avg_freight_value |
|-----|----------------|---------------------|-------------------|
| 1   | SP             | 702069.99           | 15.12             |
| 2   | RJ             | 295750.44           | 20.91             |
| 3   | MG             | 266409.84           | 20.63             |
| 4   | RS             | 132575.32           | 21.61             |
| 5   | PR             | 115645.29           | 20.47             |
| 6   | BA             | 97553.67            | 26.49             |
| 7   | SC             | 88115.65            | 21.51             |
| 8   | PE             | 57082.56            | 32.69             |
| 9   | GO             | 51375.65            | 22.56             |
| 10  | DF             | 49624.94            | 21.07             |

### Insights:

1. SP has the highest total freight cost but the lowest average freight cost per order, showing very efficient deliveries.
2. PE has the highest average freight cost per order, meaning deliveries there are expensive.
3. BA also has a high average freight cost, which could reduce profit margins.
4. RJ, MG, RS, and PR show both high total freight and average freight costs, indicating active but costlier delivery areas.
5. DF and GO have lower total freight but relatively high average freight, meaning fewer but more expensive shipments.

### Recommendation:

1. Set up warehouses or delivery hubs in high-cost states like PE and BA to reduce shipping expenses.

2. Focus more on SP for promotions and fast delivery since it has high volume and low delivery cost.
3. Offer free shipping on higher order amounts in states with expensive deliveries to encourage larger purchases.
4. Sell more digital or lightweight products in states where delivery cost is high.
5. Keep a close check on profits in states where average freight is high to avoid loss due to shipping expenses.

## 5. Analysis based on sales, freight and delivery time

1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.

Also, calculate the difference (in days) between the estimated & actual delivery date of an order.

```
SELECT
    order_id,
    customer_id,
    DATE(order_purchase_timestamp) AS purchase_date,
    DATE(order_delivered_customer_date) AS actual_delivered_date,
    DATE(order_estimated_delivery_date) AS estimated_date,
    DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY) AS
    actual_delivery_day,
    DATE_DIFF(order_delivered_customer_date, order_estimated_delivery_date, DAY)
    AS estimated_delivery_day
FROM target.orders
WHERE order_status = 'delivered';
```

| Row | order_id                       | customer_id                    | purchase_date | actual_delivered... | estimated_date | actual_delivery_day | estimated_deliver... |
|-----|--------------------------------|--------------------------------|---------------|---------------------|----------------|---------------------|----------------------|
| 1   | bfb0f9bdef84302105ad712db...   | 86dc2ffce2dfff336de2f386a78... | 2016-09-15    | 2016-11-09          | 2016-10-04     | 54                  | 36                   |
| 2   | cd3b8574c82b42fc8129f6d502...  | 7812fcebfc5e8065d31e1bb5f0...  | 2016-10-03    | 2016-10-14          | 2016-11-23     | 10                  | -39                  |
| 3   | 31b0dd6152d2e471443defb03...   | 031f08a1ebdcfb5e706831bf64...  | 2016-10-05    | 2016-10-13          | 2016-11-23     | 8                   | -40                  |
| 4   | a8c3f65a43b2e956cbfca10db4...  | c2c97a4df7be5639a8976ed8b...   | 2016-10-04    | 2016-10-20          | 2016-11-24     | 15                  | -34                  |
| 5   | a041155864e51411164582913...   | 8fbab2ff4df1a7cc9531b14a686... | 2016-10-04    | 2016-10-13          | 2016-11-24     | 9                   | -41                  |
| 6   | 50013835d7b14aefb45282586...   | 66ab9fd5f2a463cfa0995c05f4...  | 2016-10-04    | 2016-10-21          | 2016-11-24     | 17                  | -33                  |
| 7   | 1ff217aa612f6cd7c4255c9bfe9... | b3a9bf200375f53cc5c6991919...  | 2016-10-04    | 2016-10-24          | 2016-11-24     | 20                  | -30                  |
| 8   | 6eb51c1c9c69f78ea9fd4665fb...  | 7235fa173d845addc1fab54a11...  | 2016-10-04    | 2016-10-13          | 2016-11-24     | 8                   | -41                  |
| 9   | d2d67121a09535b3a7b6a000e...   | a17dc8aaea9743dbccaca067b...   | 2016-10-04    | 2016-10-13          | 2016-11-24     | 8                   | -41                  |
| 10  | 5cc475c7c03290048eb2e742c...   | 75e8f990b9e289013b1d09261...   | 2016-10-04    | 2016-12-12          | 2016-11-24     | 68                  | 18                   |

### Insights:

1. Many orders were delivered earlier than estimated, as shown by the negative values in the estimated delivery column.
2. Example: One order was delivered 41 days earlier than the estimated date — that's a huge early delivery.

3. A few orders took longer than expected, like the one with a +18 days delay, meaning it was delivered late.
4. The column actual delivery shows how many days it actually took for the order to arrive ranges between 8 to 68 days.

### Recommendation:

1. Track early deliveries to understand which shipping partners or regions are performing well.
  2. Investigate late deliveries to find reasons like poor logistics, far locations, or internal delays.
  3. Use this data to improve estimated delivery dates so that they are more accurate for customers.
  4. Prioritize customer communication when there is a risk of delay to maintain trust.
  5. Promote fast delivery options in marketing, especially in regions with consistently early deliveries.
- 
2. Find out the top 5 states with the highest & lowest average freight value.

```

WITH avg_freight_by_state AS (
  SELECT
    c.customer_state,
    ROUND(AVG(oi.freight_value), 2) AS avg_freight
  FROM
    target.orders o
  JOIN
    target.customers c ON o.customer_id = c.customer_id
  JOIN
    target.order_items oi ON o.order_id = oi.order_id
  WHERE
    o.order_status = 'delivered'
  GROUP BY
    c.customer_state
),
ranked_states AS (
  SELECT
    customer_state,
    avg_freight,
    RANK() OVER (ORDER BY avg_freight DESC) AS high_rank,
    RANK() OVER (ORDER BY avg_freight ASC) AS low_rank
  FROM
    avg_freight_by_state
)
SELECT
  customer_state,
  avg_freight,

```

```

        'Top 5 Highest' AS category
FROM
    ranked_states
WHERE
    high_rank <= 5

UNION ALL

SELECT
    customer_state,
    avg_freight,
    'Bottom 5 Lowest' AS category
FROM
    ranked_states
WHERE
    low_rank <= 5;

```

| Row | customer_state | avg_freight | category        |
|-----|----------------|-------------|-----------------|
| 1   | PR             | 20.47       | Bottom 5 Lowest |
| 2   | MG             | 20.63       | Bottom 5 Lowest |
| 3   | RJ             | 20.91       | Bottom 5 Lowest |
| 4   | DF             | 21.07       | Bottom 5 Lowest |
| 5   | SP             | 15.12       | Bottom 5 Lowest |
| 6   | RR             | 43.09       | Top 5 Highest   |
| 7   | RO             | 41.33       | Top 5 Highest   |
| 8   | PB             | 43.09       | Top 5 Highest   |
| 9   | PI             | 39.12       | Top 5 Highest   |
| 10  | AC             | 40.05       | Top 5 Highest   |

### Insights:

1. The 5 states with the lowest average freight are: SP, PR, MG, RJ, and DF.
2. The lowest freight is in SP (₹15.12), showing that deliveries here are the cheapest.
3. The 5 states with the highest average freight are: RR, RO, PB, PI, and AC.
4. The highest freight is in RR and PB (₹43.09), which is nearly 3x higher than SP.

### Recommendation:

1. Focus more on low-cost states like SP and RJ for offering free shipping and fast delivery promotions.
2. Control delivery cost in high-freight states like RR and PB by:
  - Using regional warehouses
  - Partnering with local couriers

3. In high-cost states, promote digital products or bundle orders to make shipping more efficient.
4. Track these high freight areas regularly to find ways to optimize logistics and reduce cost.

3. Find out the top 5 states with the highest & lowest average delivery time.

```
WITH delivery_time_by_state AS (  
SELECT  
    c.customer_state,  
    ROUND(AVG(DATE_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp,  
DAY)), 2) AS avg_delivery_days  
FROM  
    target.orders o  
JOIN  
    target.customers c ON o.customer_id = c.customer_id  
WHERE  
    o.order_status = 'delivered'  
GROUP BY  
    c.customer_state  
)  
,  
ranked_states AS (  
SELECT  
    customer_state,  
    avg_delivery_days,  
    RANK() OVER (ORDER BY avg_delivery_days DESC) AS high_rank,  
    RANK() OVER (ORDER BY avg_delivery_days ASC) AS low_rank  
FROM  
    delivery_time_by_state  
)  
SELECT  
    customer_state,  
    avg_delivery_days,  
    CASE  
        WHEN high_rank <= 5 THEN 'Top 5 Highest Delivery Time'  
        WHEN low_rank <= 5 THEN 'Top 5 Lowest Delivery Time'  
    END AS category  
FROM  
    ranked_states  
WHERE  
    high_rank <= 5 OR low_rank <= 5  
ORDER BY  
    category, avg_delivery_days DESC;
```

| Row | customer_state ▼ | avg_delivery_days ▼ | category ▼                  |
|-----|------------------|---------------------|-----------------------------|
| 1   | RR               | 28.98               | Top 5 Highest Delivery Time |
| 2   | AP               | 26.73               | Top 5 Highest Delivery Time |
| 3   | AM               | 25.99               | Top 5 Highest Delivery Time |
| 4   | AL               | 24.04               | Top 5 Highest Delivery Time |
| 5   | PA               | 23.32               | Top 5 Highest Delivery Time |
| 6   | SC               | 14.48               | Top 5 Lowest Delivery Time  |
| 7   | DF               | 12.51               | Top 5 Lowest Delivery Time  |
| 8   | MG               | 11.54               | Top 5 Lowest Delivery Time  |
| 9   | PR               | 11.53               | Top 5 Lowest Delivery Time  |
| 10  | SP               | 8.3                 | Top 5 Lowest Delivery Time  |

### Insights:

1. The states with the highest delivery times are RR, AP, AM, AL, and PA, with average delivery days ranging from 23 to 29 days.
2. RR has the longest delivery time (almost 29 days), showing slow delivery performance.
3. The states with the fastest delivery times are SP, PR, MG, DF, and SC, all under 15 days.
4. SP has the quickest delivery time at just 8.3 days on average.

### Recommendation:

1. Try to deliver faster in states like RR, AP, AM, AL, and PA , they are taking too long.
2. Look at fast states like SP, PR, MG, DF, and SC – learn from their delivery process.
3. Try to make delivery time similar everywhere, so all customers are happy.
4. Fast delivery means happy customers, and happy customers order more.
5. Use mobile apps or software to track packages and find quicker delivery ways.

4. Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.

You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state.

```
SELECT
    c.customer_state,
```

```

ROUND(AVG(DATE_DIFF(o.order_estimated_delivery_date,
o.order_delivered_customer_date, DAY)), 2) AS early_delivery
FROM
    target.orders o
JOIN
    target.customers c ON o.customer_id = c.customer_id
WHERE
    o.order_status = 'delivered'
GROUP BY
    c.customer_state
ORDER BY 2 desc
LIMIT 5;

```

| Row | customer_state | early_delivery |
|-----|----------------|----------------|
| 1   | AC             | 19.76          |
| 2   | RO             | 19.13          |
| 3   | AP             | 18.73          |
| 4   | AM             | 18.61          |
| 5   | RR             | 16.41          |

### Insights:

1. The top 5 states with the highest early deliveries are: AC, RO, AP, AM, and RR.
2. AC has the best performance, with orders being delivered on average about 20 days earlier than estimated.
3. All these states are showing delivery performance better than expected, with early delivery ranging between 16 to 20 days.

### Recommendation:

1. Highlight these states in marketing by offering “Faster than Expected” delivery badges.
2. Investigate what’s working well in these areas — it could be due to better logistics partners or low traffic zones.
3. Use this data to improve estimated delivery dates for better accuracy and customer trust.
4. Consider offering same-day or next-day delivery services in these regions to boost customer satisfaction even more.
5. Monitor these states to maintain the high delivery performance and identify any upcoming changes.

### 6. Analysis based on the payments:

1. Find the month on month no. of orders placed using different payment types.

```
SELECT payment_type, a.month, count(order_id) as total_orders
FROM (SELECT *,extract(month from order_purchase_timestamp) as month
FROM target.orders inner join target.payments using(order_id)) a
group by 1,2
order by 1,2;
```

| Row | payment_type ▼ | month ▼ | total_orders ▼ |
|-----|----------------|---------|----------------|
| 1   | UPI            | 1       | 1715           |
| 2   | UPI            | 2       | 1723           |
| 3   | UPI            | 3       | 1942           |
| 4   | UPI            | 4       | 1783           |
| 5   | UPI            | 5       | 2035           |
| 6   | UPI            | 6       | 1807           |
| 7   | UPI            | 7       | 2074           |
| 8   | UPI            | 8       | 2077           |
| 9   | UPI            | 9       | 903            |
| 10  | UPI            | 10      | 1056           |

### Insights:

1. UPI orders were consistently increasing from January (1715) to August (2077).
2. The highest number of UPI orders was in August (2077 orders).
3. There is a sharp drop in orders after August, with only 903 in September and 1056 in October.
4. This suggests a strong UPI trend in the first 8 months, followed by a decline.

### Recommendation:

1. Focus UPI-based offers and promotions during January to August, when customer activity is high.
  2. Investigate the drop after August — could be due to seasonality, technical issues, or shifting customer preferences.
  3. Run festive campaigns or cashback offers in later months to revive UPI usage.
  4. Monitor UPI usage trends monthly to plan better for marketing, payment options, and inventory.
  5. Encourage UPI payments with discounts or fast checkout features, especially in peak months.
2. Find the no. of orders placed on the basis of the payment installments that have been paid.



```

SELECT
    count(DISTINCT o.order_id) as total_order,
    p.payment_installments
FROM
    target.orders o
JOIN
    target.payments p
ON
    o.order_id=p.order_id
WHERE payment_installments >= 1
GROUP BY payment_installments
ORDER BY 2;

```

| Row | total_order | payment_installments |
|-----|-------------|----------------------|
| 1   | 49060       | 1                    |
| 2   | 12389       | 2                    |
| 3   | 10443       | 3                    |
| 4   | 7088        | 4                    |
| 5   | 5234        | 5                    |
| 6   | 3916        | 6                    |
| 7   | 1623        | 7                    |
| 8   | 4253        | 8                    |
| 9   | 644         | 9                    |
| 10  | 5315        | 10                   |

### Insights:

1. Most customers prefer to pay in 1 installment — it has the highest order count (49,060 orders).
2. The number of orders decreases as the number of installments increases.
3. Installments 7 to 9 are the least used, with very low order counts.
4. Installment 10 has a slight rise (5,315 orders) — possibly due to EMI offers

### Recommendation:

1. Focus on promoting 1 and 2-installment options, as they are the most preferred.
2. Provide special EMI offers only on selected products to encourage longer installment options (like 6–10 months).

3. Investigate why installments 7–9 are unpopular — may need better interest rates or awareness.
4. Use this data to design custom payment plans — for example, recommend shorter installment options for low-value products.
5. Promote 10-month EMI during high-value product campaigns, since there's a demand spike there.